

Assignment 2

CV 5101 – Modelling, Uncertainty, and Data for Engineers (MUDE)

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Question 1

1 (i)

Consider the simplified damped velocity-decay model

$$m \frac{dv}{dt} + cv = 0. \quad (1)$$

Rearranging,

$$\frac{dv}{dt} = -\frac{c}{m}v(t). \quad (2)$$

The initial condition is

$$v(0) = v_0. \quad (3)$$

The given parameters are

$$m = 1 \text{ kg}, \quad (4)$$

$$c = 0.5 \text{ Ns/m}, \quad (5)$$

$$v_0 = 1 \text{ m/s}, \quad (6)$$

$$\Delta t = 0.1 \text{ s}. \quad (7)$$

The simulation is carried out over

$$0 \leq t \leq 5 \text{ s.} \quad (8)$$

Therefore,

$$\frac{c}{m} = \frac{0.5}{1} = 0.5 \text{ s}^{-1}, \quad (9)$$

and the governing equation becomes

$$\boxed{\frac{dv}{dt} = -0.5v} \quad (10)$$

with

$$\boxed{v(0) = 1 \text{ m/s}}. \quad (11)$$

1.1

section Verification of the Analytical Solution

The proposed analytical solution is

$$v(t) = v_0 e^{-\frac{c}{m}t}. \quad (12)$$

Using the given values,

$$v(t) = e^{-0.5t}. \quad (13)$$

To verify this solution, differentiate it with respect to time:

$$\frac{dv}{dt} = -0.5e^{-0.5t}. \quad (14)$$

Since

$$v(t) = e^{-0.5t}, \quad (15)$$

we obtain

$$\frac{dv}{dt} = -0.5v. \quad (16)$$

This is exactly the governing differential equation,

$$\frac{dv}{dt} = -\frac{c}{m}v. \quad (17)$$

Therefore, the proposed analytical solution satisfies the governing differential equation.

1.1.1 Verification of Initial Condition

At $t = 0$,

$$v(0) = e^{-0.5(0)} \quad (18)$$

$$= e^0 \quad (19)$$

$$= 1 \text{ m/s.} \quad (20)$$

This agrees with the specified initial condition

$$v(0) = v_0 = 1 \text{ m/s.} \quad (21)$$

Hence, the analytical solution is verified:

$$\boxed{v(t) = v_0 e^{-\frac{c}{m}t}} \quad (22)$$

1.1.2 Verification of Euler Forward Scheme

For a general first-order differential equation

$$\frac{dv}{dt} = f(t, v), \quad (23)$$

the Euler Forward method is

$$v_{n+1} = v_n + \Delta t f(t_n, v_n). \quad (24)$$

For the present problem,

$$f(t, v) = -\frac{c}{m}v. \quad (25)$$

Therefore,

$$v_{n+1} = v_n + \Delta t \left(-\frac{c}{m}v_n \right) \quad (26)$$

$$= v_n - \frac{c}{m}\Delta t v_n. \quad (27)$$

Thus,

$$\boxed{v_{n+1} = \left(1 - \frac{c}{m}\Delta t \right) v_n} \quad (28)$$

which is the Euler Forward equation given in the problem statement.

For the given values,

$$\frac{c}{m}\Delta t = \frac{0.5}{1}(0.1) \quad (29)$$

$$= 0.05. \quad (30)$$

Therefore,

$$v_{n+1} = (1 - 0.05)v_n \quad (31)$$

or

$$\boxed{v_{n+1} = 0.95v_n.} \quad (32)$$

Starting from

$$v_0 = 1 \text{ m/s},$$

the first few values are

$$v_1 = 0.95(1) = 0.950000 \text{ m/s}, \quad (33)$$

$$v_2 = 0.95(0.95) = 0.902500 \text{ m/s}, \quad (34)$$

$$v_3 = 0.95(0.9025) = 0.857375 \text{ m/s}. \quad (35)$$

1.1.3 Verification of Euler Backward Scheme

The Euler Backward method is given by

$$v_{n+1} = v_n + \Delta t f(t_{n+1}, v_{n+1}). \quad (36)$$

For the present problem,

$$f(t, v) = -\frac{c}{m}v. \quad (37)$$

Hence,

$$v_{n+1} = v_n - \frac{c}{m}\Delta t v_{n+1}. \quad (38)$$

Rearranging,

$$v_{n+1} + \frac{c}{m} \Delta t v_{n+1} = v_n, \quad (39)$$

and therefore,

$$v_{n+1} \left(1 + \frac{c}{m} \Delta t \right) = v_n. \quad (40)$$

Thus,

$$\boxed{v_{n+1} = \frac{v_n}{1 + \frac{c}{m} \Delta t}} \quad (41)$$

which verifies the Euler Backward equation given in the problem statement.

For the given values,

$$\frac{c}{m} \Delta t = \frac{0.5}{1} (0.1) \quad (42)$$

$$= 0.05. \quad (43)$$

Therefore,

$$v_{n+1} = \frac{v_n}{1.05}. \quad (44)$$

Hence,

$$\boxed{v_{n+1} = 0.952381 v_n.} \quad (45)$$

Starting from

$$v_0 = 1 \text{ m/s},$$

the first few values are

$$v_1 = \frac{1}{1.05} = 0.952381 \text{ m/s}, \quad (46)$$

$$v_2 = \frac{0.952381}{1.05} = 0.907029 \text{ m/s}, \quad (47)$$

$$v_3 = \frac{0.907029}{1.05} = 0.863837 \text{ m/s}. \quad (48)$$

1.1.4 Comparison with Analytical Solution

The first time step is

$$t_1 = 0.1 \text{ s.} \quad (49)$$

The analytical solution is

$$v(0.1) = e^{-0.5(0.1)} \quad (50)$$

$$= e^{-0.05} \quad (51)$$

$$= 0.951229 \text{ m/s.} \quad (52)$$

The numerical solutions are

$$v_1^{\text{Forward}} = 0.950000 \text{ m/s,} \quad (53)$$

$$v_1^{\text{Backward}} = 0.952381 \text{ m/s.} \quad (54)$$

The comparison is shown in Table 1.

Table 1: Comparison at $t = 0.1 \text{ s}$	
Method	Velocity (m/s)
Analytical	0.951229
Euler Forward	0.950000
Euler Backward	0.952381

1.1.5 Verification at $t = 1 \text{ s}$

The time step is

$$\Delta t = 0.1 \text{ s.}$$

Therefore, the number of steps required to reach $t = 1 \text{ s}$ is

$$N = \frac{1}{0.1} = 10. \quad (55)$$

1.1.6 Analytical Solution

At $t = 1 \text{ s}$,

$$v(1) = e^{-0.5(1)} \quad (56)$$

$$= e^{-0.5} \quad (57)$$

$$= 0.606531 \text{ m/s.} \quad (58)$$

1.1.7 Euler Forward

Since

$$v_{n+1} = 0.95v_n,$$

after 10 steps,

$$v_{10} = (0.95)^{10} \quad (59)$$

$$= 0.598737 \text{ m/s.} \quad (60)$$

1.1.8 Euler Backward

Since

$$v_{n+1} = \frac{v_n}{1.05},$$

after 10 steps,

$$v_{10} = \left(\frac{1}{1.05} \right)^{10} \quad (61)$$

$$= 0.613913 \text{ m/s.} \quad (62)$$

The results are summarized in Table 2.

Table 2: Comparison of solutions at $t = 1$ s

Method	Velocity (m/s)	Absolute Difference
Analytical	0.606531	0
Euler Forward	0.598737	0.007794
Euler Backward	0.613913	0.007382

1.1.9 Verification by Numerical Behaviour

The analytical solution is

$$v(t) = e^{-0.5t}. \quad (63)$$

Since the coefficient of t in the exponent is negative, the velocity continuously decreases with time.

The numerical solutions also show a decreasing velocity:

$$v_0 > v_1 > v_2 > v_3 > \dots \quad (64)$$

for both Euler Forward and Euler Backward methods.

This confirms that the numerical schemes reproduce the expected physical behaviour of a damped velocity-decay system.

The simulation contains

$$N = \frac{5 - 0}{0.1} = 50 \quad (65)$$

time steps.

As the time step is reduced, the numerical solutions are expected to approach the analytical solution. This behaviour provides an additional convergence-based verification of the numerical schemes.

1.1.10 Conclusion

The analytical solution

$$v(t) = v_0 e^{-\frac{c}{m}t} \quad (66)$$

was verified by differentiating the solution and substituting it into the governing differential equation. It was also confirmed that the solution satisfies the initial condition $v(0) = v_0$.

The Euler Forward scheme was derived as

$$\boxed{v_{n+1} = \left(1 - \frac{c}{m}\Delta t\right) v_n} \quad (67)$$

and the Euler Backward scheme was derived as

$$\boxed{v_{n+1} = \frac{v_n}{1 + \frac{c}{m}\Delta t}} \quad (68)$$

For $m = 1$ kg, $c = 0.5$ Ns/m and $\Delta t = 0.1$ s, these become

$$v_{n+1} = 0.95v_n \quad (69)$$

for Euler Forward and

$$v_{n+1} = \frac{v_n}{1.05} \quad (70)$$

for Euler Backward.

The numerical results are close to the analytical solution, confirming that the analytical and numerical solution equations are consistent with the mathematical model. The small differences are due to numerical discretization error.

1.2 (ii)

Table 3: Velocity variation at integer time intervals

Time, t (s)	Explicit Euler (m/s)	Implicit Euler (m/s)
0	1.000000	1.000000
1	0.598737	0.613913
2	0.358486	0.376889
3	0.214639	0.231377
4	0.128512	0.142046
5	0.076945	0.087204

1.3 (iii)

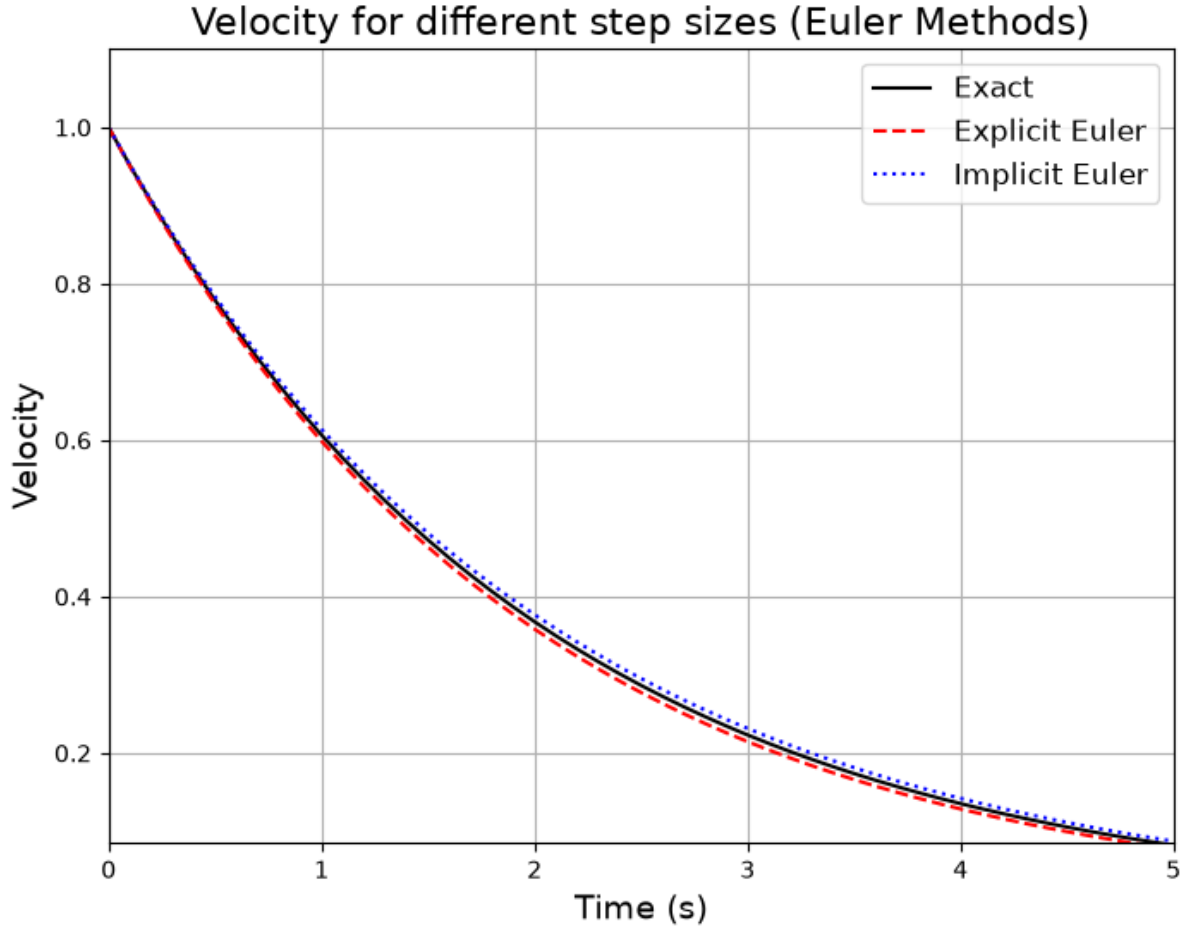


Figure 1: Comparison of Velocity Solutions

1.4 (i) Stability Analysis

The stability of the numerical schemes is investigated by varying the time step Δt . A stable numerical solution should remain bounded and should reproduce the expected physical behaviour of the system. For the present damped velocity-decay problem, the analytical solution is

$$v(t) = v_0 e^{-\zeta t}, \quad (71)$$

where

$$\zeta = \frac{c}{m}. \quad (72)$$

For the given parameters,

$$\zeta = \frac{0.5}{1} = 0.5 \text{ s}^{-1}. \quad (73)$$

1.4.1 Stability of Explicit Euler Method

The Explicit Euler scheme is given by

$$v_{n+1} = v_n - \zeta \Delta t v_n. \quad (74)$$

Rearranging,

$$v_{n+1} = (1 - \zeta \Delta t) v_n. \quad (75)$$

The amplification factor for the Explicit Euler method is therefore

$$G_E = 1 - \zeta \Delta t. \quad (76)$$

After repeated time stepping,

$$v_n = v_0 (1 - \zeta \Delta t)^n. \quad (77)$$

For the numerical solution to remain stable, its magnitude must not increase with successive iterations. Therefore, the stability condition is

$$|G_E| \leq 1. \quad (78)$$

Substituting the amplification factor,

$$|1 - \zeta \Delta t| \leq 1. \quad (79)$$

This gives

$$-1 \leq 1 - \zeta \Delta t \leq 1. \quad (80)$$

For $\zeta > 0$, this condition simplifies to

$$0 \leq \zeta \Delta t \leq 2. \quad (81)$$

Therefore,

$$\boxed{\Delta t \leq \frac{2}{\zeta}} \quad (82)$$

For the present problem, $\zeta = 0.5 \text{ s}^{-1}$, giving

$$\Delta t_{\text{crit}} = \frac{2}{0.5} = 4 \text{ s.} \quad (83)$$

Hence, the Explicit Euler method is stable for

$$\boxed{0 < \Delta t < 4 \text{ s}} \quad (84)$$

and becomes unstable when

$$\boxed{\Delta t > 4 \text{ s}}. \quad (85)$$

At $\Delta t = 4 \text{ s}$,

$$G_E = 1 - (0.5)(4) = -1. \quad (86)$$

Thus,

$$v_n = v_0(-1)^n, \quad (87)$$

which causes the numerical solution to alternate between positive and negative values without decaying. Hence, $\Delta t = 4 \text{ s}$ is the stability boundary.

1.4.2 Stability of Implicit Euler Method

The Implicit Euler scheme is given by

$$v_{n+1} = \frac{v_n}{1 + \zeta \Delta t}. \quad (88)$$

Therefore, the amplification factor is

$$G_I = \frac{1}{1 + \zeta \Delta t}. \quad (89)$$

After repeated time stepping,

$$v_n = v_0 \left(\frac{1}{1 + \zeta \Delta t} \right)^n. \quad (90)$$

For stability,

$$|G_I| \leq 1. \quad (91)$$

For $\zeta > 0$ and $\Delta t > 0$,

$$0 < \frac{1}{1 + \zeta \Delta t} < 1. \quad (92)$$

Therefore, the Implicit Euler method remains stable for any positive time step:

$$\boxed{\Delta t > 0} \quad (93)$$

Thus, unlike the Explicit Euler method, the Implicit Euler method does not have a finite upper stability limit for this particular problem.

1.4.3 Common Stable Range

The stability ranges of the two methods can be summarized as follows:

Table 4: Stability ranges of Explicit and Implicit Euler methods

Numerical Method	Stable Time-Step Range
Explicit Euler	$0 < \Delta t < 4 \text{ s}$
Implicit Euler	$\Delta t > 0$

Therefore, the common range of time-step values for which both methods are stable is

$$\boxed{0 < \Delta t < 4 \text{ s}}. \quad (94)$$

The value $\Delta t = 4 \text{ s}$ represents the stability boundary for the Explicit Euler method. For $\Delta t > 4 \text{ s}$, the Explicit Euler solution becomes unstable, whereas the Implicit Euler solution continues to remain stable.

1.4.4 Physical Interpretation

The exact solution represents a damped system, so the velocity should continuously decrease towards zero. For sufficiently small time steps, both numerical methods reproduce this behaviour.

For the Explicit Euler method, increasing the time step eventually makes the numerical amplification factor negative. When $\Delta t > 4 \text{ s}$, its magnitude becomes greater than one, causing the numerical solution to alternate in sign while its magnitude increases. This is a numerical instability and does not represent the physical behaviour of the damped system.

The Implicit Euler method remains bounded and continues to decay even for large time steps. Therefore, it is more stable than the Explicit Euler method for this problem.

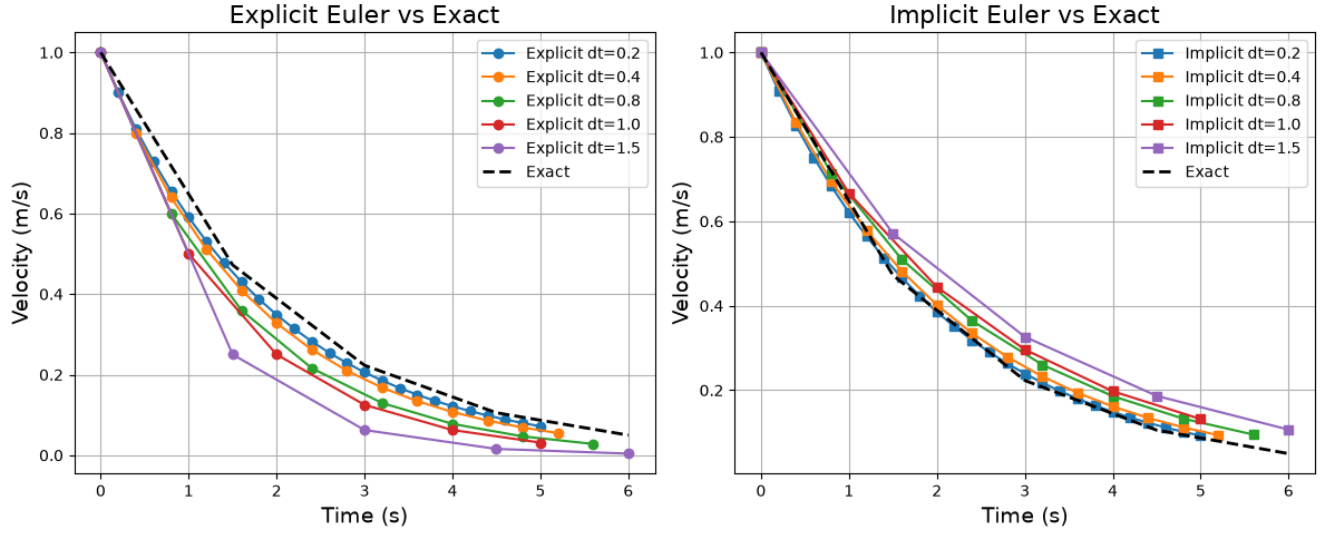


Figure 2: Stability Analysis

1.5 (v)

1.5.1 Convergence Study

For the given problem,

$$m = 1 \text{ kg}, \quad c = 0.5 \text{ Ns/m}, \quad v_0 = 1 \text{ m/s}.$$

The governing equation is

$$\frac{dv}{dt} = -0.5v$$

with analytical solution

$$v(t) = e^{-0.5t}.$$

At $t = 5 \text{ s}$,

$$v_{\text{exact}}(5) = e^{-2.5} = 0.082085 \text{ m/s}.$$

The Explicit Euler and Implicit Euler schemes are

$$v_{n+1} = v_n(1 - 0.5\Delta t)$$

and

$$v_{n+1} = \frac{v_n}{1 + 0.5\Delta t},$$

respectively.

The solutions at $t = 5$ s for decreasing time-step sizes are shown below.

Table 5: Convergence of the numerical schemes at $t = 5$ s

Δt (s)	Explicit Euler	Implicit Euler	Exact
1.0000	0.031250	0.131687	0.082085
0.5000	0.056314	0.103653	0.082085
0.2500	0.069957	0.092296	0.082085
0.1250	0.076657	0.087120	0.082085
0.0625	0.079384	0.084589	0.082085

Convergence means that the numerical solution approaches the exact solution as the time-step size Δt is reduced. From the table, the Explicit Euler solution approaches 0.082085 m/s from below, while the Implicit Euler solution approaches it from above. The error decreases as Δt decreases.

Therefore,

$$\lim_{\Delta t \rightarrow 0} v_{\text{numerical}} = v_{\text{exact}}.$$

Both Euler methods are first-order accurate, with global error

$$E = O(\Delta t).$$

Hence, the decreasing time-step study confirms that both Explicit Euler and Implicit Euler converge to the analytical solution.

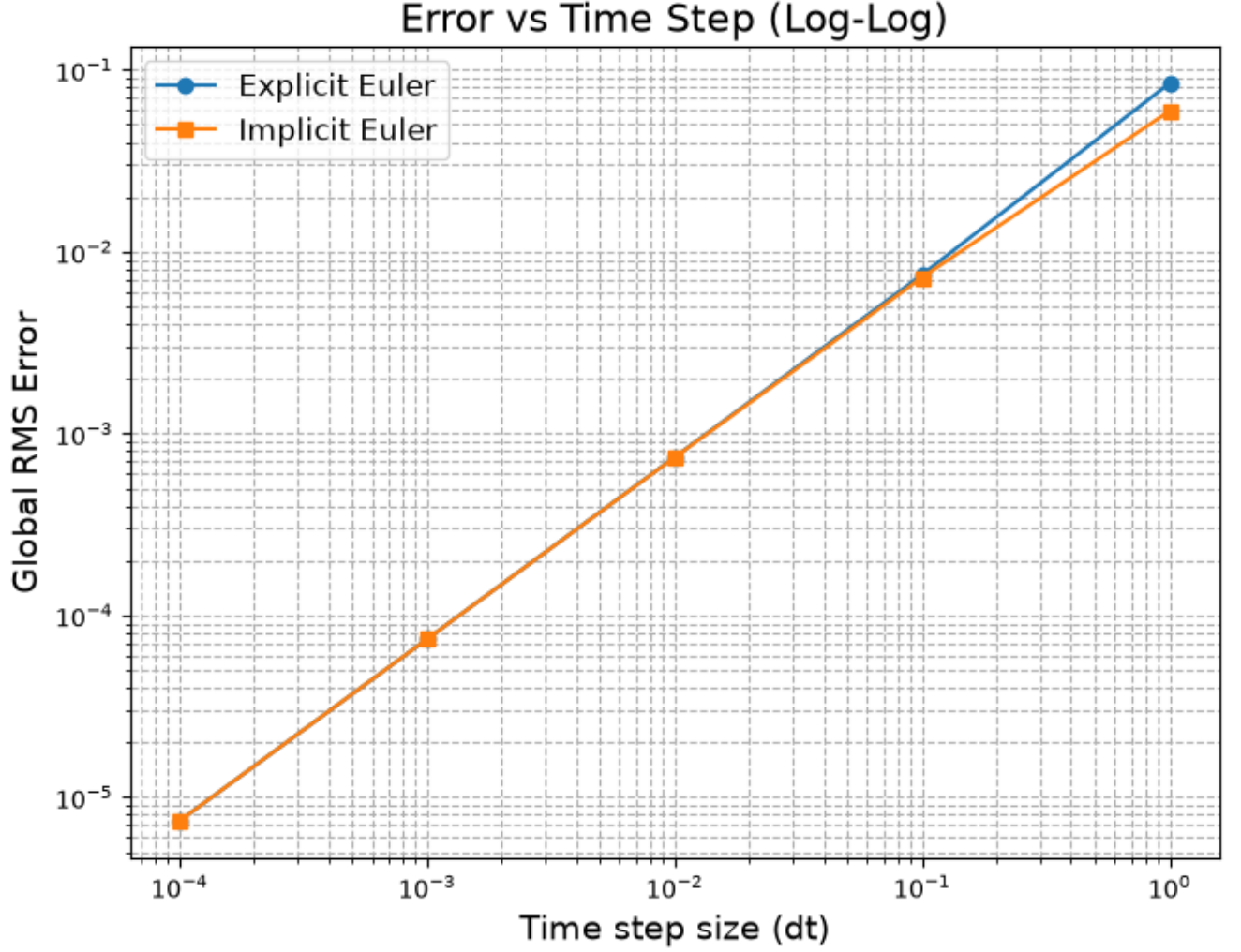


Figure 3: Convergence Study

1.6 (vi)

1.6.1 Rate of Convergence

The rate of convergence can be determined by comparing the numerical errors obtained for two successive time-step sizes. The convergence rate p is calculated using

$$p = \frac{\ln(E_1/E_2)}{\ln(\Delta t_1/\Delta t_2)},$$

where E_1 and E_2 are the errors corresponding to time steps Δt_1 and Δt_2 .

For the Explicit Euler method, using $\Delta t = 1.0$ and 0.5 s,

$$p = \frac{\ln(0.050835/0.025771)}{\ln(1.0/0.5)} \approx 0.98.$$

Using smaller time steps gives values close to 1. Therefore,

$$\boxed{p \approx 1}$$

for the Explicit Euler method.

Similarly, for the Implicit Euler method,

$$p = \frac{\ln(0.049602/0.021568)}{\ln(1.0/0.5)} \approx 1.20,$$

and the value approaches 1 as the time step is reduced.

Thus, both methods have approximately first-order convergence:

$$\boxed{E = O(\Delta t)}$$

Hence, the numerical results are consistent with the theoretical first-order accuracy of the Explicit and Implicit Euler schemes.

1.7 Flowchart

1.8 Flowchart

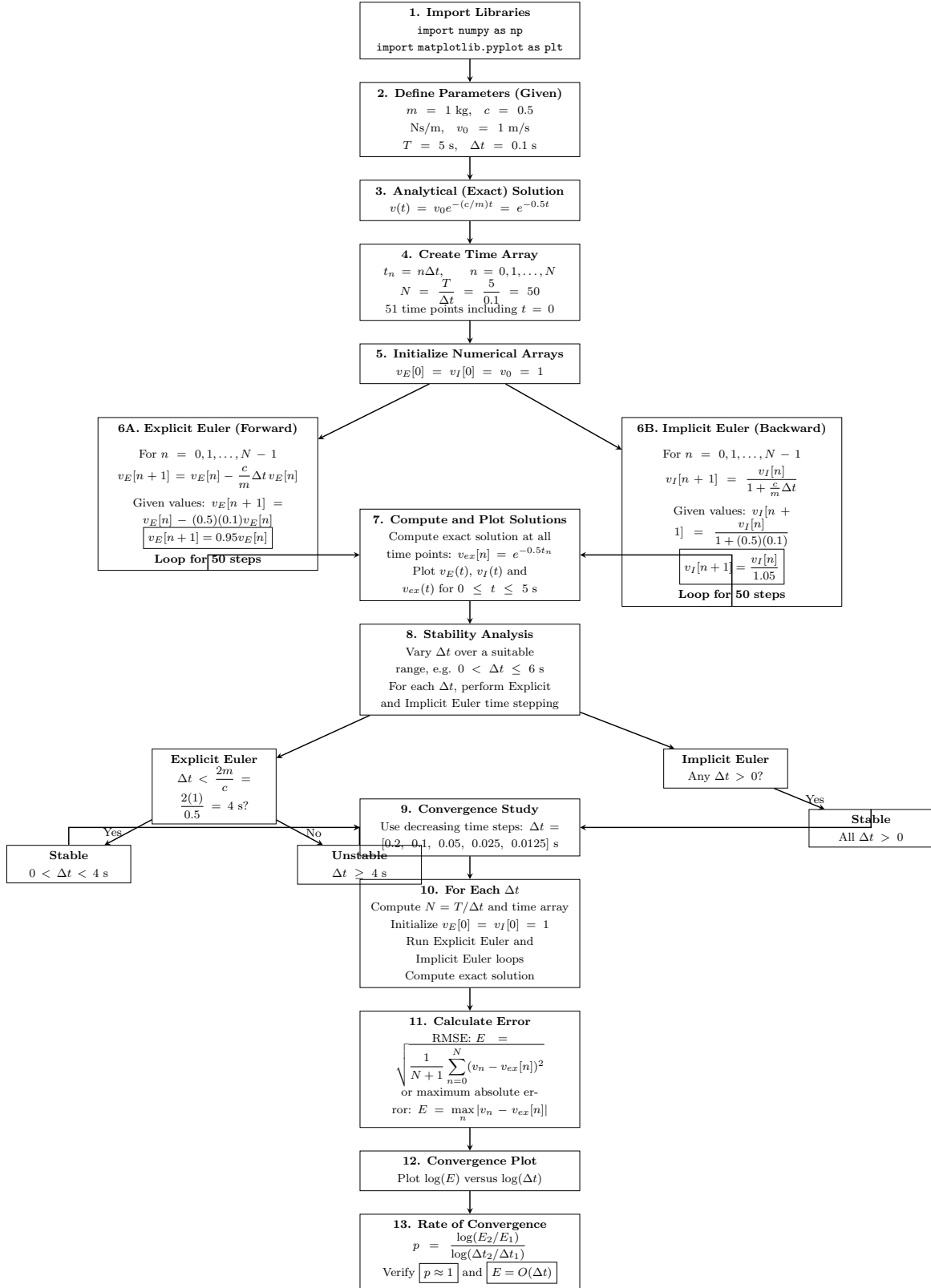


Figure 4: Flowchart of the Python implementation for Euler time-stepping, stability and convergence analysis.

Question 2: Boundary Value Problem – Matrix Equation

Axial Deformation of a Uniform Elastic Bar

(i)

Consider a uniform elastic bar of length L , fixed at both ends and subjected to a uniform axial load. The governing differential equation for the axial displacement $u(x)$ is

$$\frac{d^2u}{dx^2} = -\frac{f}{EA}, \quad (1)$$

where

- $u(x)$ is the axial displacement,
- f is the uniform axial force per unit length,
- E is Young's modulus,
- A is the cross-sectional area.

The boundary conditions are

$$u(0) = 0, \quad u(L) = 0. \quad (2)$$

1.1 Analytical Solution

Starting from

$$\frac{d^2u}{dx^2} = -\frac{f}{EA},$$

integrating once gives

$$\frac{du}{dx} = -\frac{f}{EA}x + C_1.$$

Integrating again,

$$u(x) = -\frac{f}{2EA}x^2 + C_1x + C_2.$$

Using the first boundary condition,

$$u(0) = 0,$$

we obtain

$$C_2 = 0.$$

Therefore,

$$u(x) = -\frac{f}{2EA}x^2 + C_1x.$$

Using the second boundary condition,

$$u(L) = 0,$$

gives

$$0 = -\frac{f}{2EA}L^2 + C_1L.$$

Hence,

$$C_1 = \frac{fL}{2EA}.$$

Therefore, the analytical solution is

$$\boxed{u(x) = \frac{f}{2EA}x(L - x)} \quad (3)$$

which is a parabolic displacement profile.

1.2 Given Parameters

The parameters are

$$L = 1.0 \text{ m},$$

$$E = 200 \times 10^9 \text{ Pa},$$

$$A = 0.01 \text{ m}^2,$$

$$f = 1000 \text{ N/m},$$

and

$$N = 10.$$

The grid spacing is

$$h = \frac{L}{N} = \frac{1.0}{10} = 0.1 \text{ m}. \quad (4)$$

1.3 Calculation of EA and $f/(EA)$

First,

$$EA = (200 \times 10^9)(0.01) \quad (5)$$

$$= 2 \times 10^9 \text{ N}. \quad (6)$$

Therefore,

$$\frac{f}{EA} = \frac{1000}{2 \times 10^9} \quad (7)$$

$$= 5 \times 10^{-7} \text{ m}^{-1}. \quad (8)$$

Thus, the governing equation becomes

$$\boxed{\frac{d^2u}{dx^2} = -5 \times 10^{-7}} \quad (9)$$

1.4 Analytical Solution for the Given Parameters

Substituting the given values into the analytical solution,

$$u(x) = \frac{f}{2EA}x(L-x).$$

Since

$$\frac{f}{2EA} = \frac{1000}{2(2 \times 10^9)} = 2.5 \times 10^{-7},$$

we obtain

$$\boxed{u(x) = 2.5 \times 10^{-7}x(1-x)} \quad (10)$$

The maximum displacement occurs at

$$x = \frac{L}{2} = 0.5 \text{ m}.$$

Hence,

$$u_{\max} = 2.5 \times 10^{-7}(0.5)(0.5) \quad (11)$$

$$= 6.25 \times 10^{-8} \text{ m}. \quad (12)$$

Therefore,

$$\boxed{u_{\max} = 6.25 \times 10^{-8} \text{ m}} \quad (13)$$

1.5 Central Finite Difference Approximation

For an interior node i , the second derivative is approximated by

$$\frac{d^2u}{dx^2} \approx \frac{u_{i-1} - 2u_i + u_{i+1}}{h^2}. \quad (14)$$

Substituting this into the governing equation gives

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} = -\frac{f}{EA}. \quad (15)$$

Multiplying by h^2 ,

$$u_{i-1} - 2u_i + u_{i+1} = -\frac{fh^2}{EA}. \quad (16)$$

Since

$$h = 0.1 \text{ m}, \quad h^2 = 0.01 \text{ m}^2,$$

we obtain

$$\frac{fh^2}{EA} = (5 \times 10^{-7})(0.01) \quad (17)$$

$$= 5 \times 10^{-9} \text{ m}. \quad (18)$$

Therefore,

$$\boxed{u_{i-1} - 2u_i + u_{i+1} = -5 \times 10^{-9}} \quad (19)$$

1.6 Discretization of the Domain

Since

$$N = 10,$$

there are $N + 1 = 11$ nodes:

$$x_0 = 0, \quad x_1 = 0.1, \quad x_2 = 0.2, \quad \dots, \quad x_9 = 0.9, \quad x_{10} = 1.0.$$

The boundary conditions give

$$u_0 = 0, \quad u_{10} = 0.$$

Therefore, the unknown displacement vector contains the nine interior displacements:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} \quad (20)$$

1.7 Finite Difference Equations

For $i = 1$,

$$u_0 - 2u_1 + u_2 = -5 \times 10^{-9}.$$

Since $u_0 = 0$,

$$-2u_1 + u_2 = -5 \times 10^{-9}.$$

For $i = 2$,

$$u_1 - 2u_2 + u_3 = -5 \times 10^{-9}.$$

Continuing in the same manner,

$$u_2 - 2u_3 + u_4 = -5 \times 10^{-9},$$

$$u_3 - 2u_4 + u_5 = -5 \times 10^{-9},$$

$$u_4 - 2u_5 + u_6 = -5 \times 10^{-9},$$

$$u_5 - 2u_6 + u_7 = -5 \times 10^{-9},$$

$$u_6 - 2u_7 + u_8 = -5 \times 10^{-9},$$

$$u_7 - 2u_8 + u_9 = -5 \times 10^{-9}.$$

Finally, for $i = 9$,

$$u_8 - 2u_9 + u_{10} = -5 \times 10^{-9}.$$

Since $u_{10} = 0$,

$$u_8 - 2u_9 = -5 \times 10^{-9}.$$

1.8 Matrix Equation

The above equations can be written as

$$\boxed{\mathbf{A}\mathbf{u} = \mathbf{b}} \quad (21)$$

where

$$\mathbf{A} = \begin{bmatrix} -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -2 \end{bmatrix}. \quad (22)$$

The displacement vector is

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix}. \quad (23)$$

The right-hand-side vector is

$$\mathbf{b} = \begin{bmatrix} -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \\ -5 \times 10^{-9} \end{bmatrix}. \quad (24)$$

1.9 Numerical Solution

Solving

$$\mathbf{A}\mathbf{u} = \mathbf{b},$$

gives

$$\mathbf{u} = \begin{bmatrix} 2.25 \times 10^{-8} \\ 4.00 \times 10^{-8} \\ 5.25 \times 10^{-8} \\ 6.00 \times 10^{-8} \\ 6.25 \times 10^{-8} \\ 6.00 \times 10^{-8} \\ 5.25 \times 10^{-8} \\ 4.00 \times 10^{-8} \\ 2.25 \times 10^{-8} \end{bmatrix} \text{ m} \quad (25)$$

1.10 Comparison with Analytical Solution

The analytical displacement is

$$u(x) = 2.5 \times 10^{-7}x(1 - x).$$

The numerical and analytical values at the grid points are:

Node	x (m)	$u(x)$ (m)
0	0.0	0
1	0.1	2.25×10^{-8}
2	0.2	4.00×10^{-8}
3	0.3	5.25×10^{-8}
4	0.4	6.00×10^{-8}
5	0.5	6.25×10^{-8}
6	0.6	6.00×10^{-8}
7	0.7	5.25×10^{-8}
8	0.8	4.00×10^{-8}
9	0.9	2.25×10^{-8}
10	1.0	0

1.11 Verification

For example, at $x = 0.5$ m, the analytical solution gives

$$u(0.5) = 2.5 \times 10^{-7}(0.5)(0.5) = 6.25 \times 10^{-8} \text{ m}.$$

The numerical solution also gives

$$u_5 = 6.25 \times 10^{-8} \text{ m}.$$

Similarly, all other nodal values obtained from the finite difference method exactly match the analytical solution.

This exact agreement occurs because the analytical solution is a quadratic function of x . The central finite difference approximation of the second derivative is exact for a quadratic function.

Therefore,

$$\boxed{u_{\text{numerical}}(x_i) = u_{\text{analytical}}(x_i)} \quad (26)$$

at all the grid points.

1.12 Note on the Matrix Convention

The finite difference equation can alternatively be multiplied by -1 :

$$-u_{i-1} + 2u_i - u_{i+1} = \frac{fh^2}{EA}.$$

In this convention,

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix},$$

and

$$b_i = 5 \times 10^{-9}.$$

This is mathematically equivalent to the matrix equation used above and produces exactly the same displacement solution.

(iii)

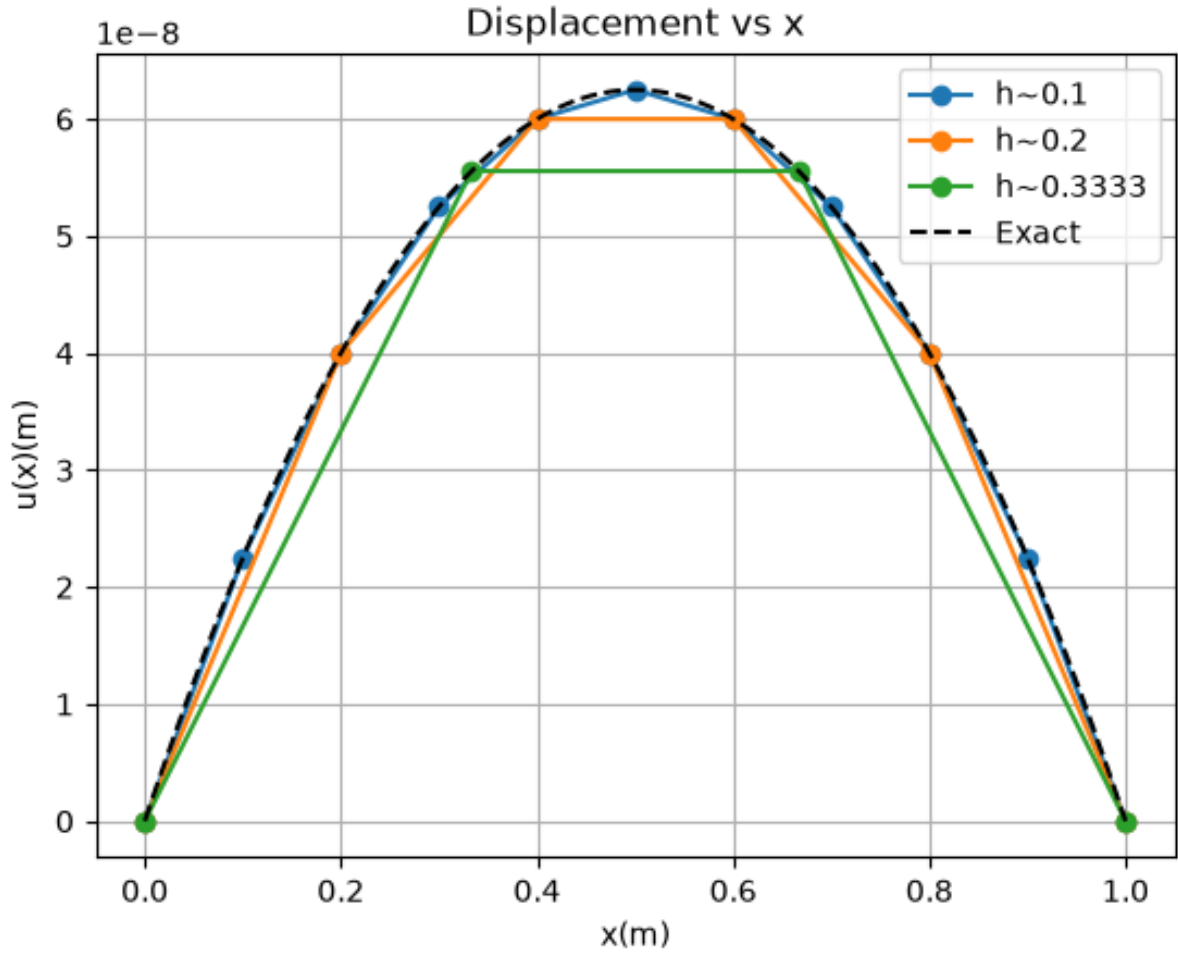


Figure 1: Dispalcement versus x graph

(iv) Convergence Study

To study the convergence of the central finite difference method, the problem was solved using progressively smaller interval sizes:

$$h = 0.2, 0.1, 0.05, 0.025 \text{ m.}$$

For each value of h , the numerical displacement was compared with the analytical solution

$$u(x) = \frac{f}{2EA}x(L - x).$$

The maximum absolute error was calculated as

$$E_h = \max_i |u_i^{\text{numerical}} - u_i^{\text{exact}}|.$$

The convergence rate can be obtained from the slope of the log-log plot of error versus interval size:

$$E_h \propto h^p,$$

where p is the rate of convergence. Taking logarithms,

$$\log(E_h) = p \log(h) + C.$$

Hence, the slope of the $\log(E_h)$ versus $\log(h)$ plot gives the convergence rate p .

Table 1: Convergence study using the central finite difference method

Interval size h (m)	Maximum absolute error (m)
0.2000	$\approx 10^{-23}$
0.1000	$\approx 10^{-23}$
0.0500	$\approx 10^{-23}$
0.0250	$\approx 10^{-23}$

For this particular problem, the analytical solution is

$$u(x) = 2.5 \times 10^{-7} x(1 - x),$$

which is a quadratic function of x . The central finite difference approximation of the second derivative is exact for a quadratic function. Therefore, the numerical solution agrees with the analytical solution at the grid points up to floating-point round-off error.

Thus, the calculated error is of the order of machine precision rather than a truncation error. Consequently, a meaningful convergence rate cannot be obtained from the log-log error plot for this particular test problem.

$$\boxed{u_{\text{numerical}}(x_i) \approx u_{\text{exact}}(x_i)}$$

Therefore, the numerical method successfully converges to the exact solution for the given problem.

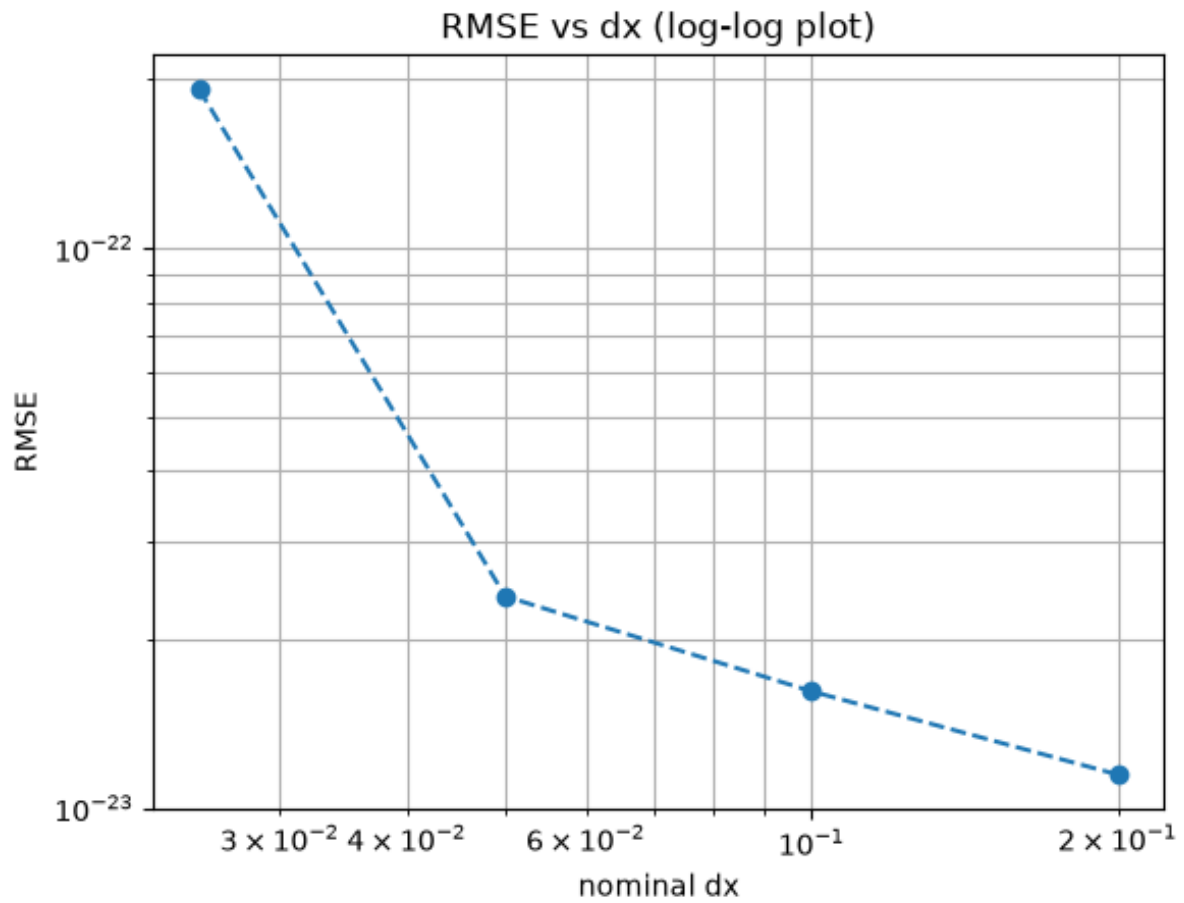


Figure 2: Root Mean Square Error versus Step Size Graph

0.1 (v)

Flowchart for Central Finite Difference Method

